

Problem 1. Averaging argument.

In the lecture, we have seen the following version of the “No free lunch” theorem:

Theorem 1. *Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that*

$$\mathcal{R}_\star = 0, \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq \frac{1}{4}.$$

You will prove the following corollary using [Theorem 1](#):

Corollary 2. *Let $\mathcal{X}$ be infinite and let $\mathcal{H}$ be the set of all functions from $\mathcal{X}$ to $\{0, 1\}$. Then $\mathcal{H}$ is not PAC-learnable.*

Solution.

Problem 2. ERM for learning closed intervals.

Let $\mathcal{X} = \mathbb{R}$ and $\mathcal{H}$ be the set of indicator functions of closed intervals and the empty set:

$$\mathcal{H} = \{\mathbf{1}_{a \leq x \leq b} : \forall a, b \in \mathbb{R}\}.$$

Assume binary classification with 0–1 loss and realizability. Prove that $\mathcal{H}$ can be PAC-learned using ERM with sample complexity

$$O\left(\frac{\log(1/\delta)}{\epsilon}\right).$$

You need to provide a description of the learning algorithm and prove its sample complexity and the PAC learning guarantee.

Solution.

Problem 3. VC dimensions.

Given some finite domain set, $\mathcal{X}$, and a number $k \leq |\mathcal{X}|$, figure out the VC dimension of each of the following classes (and prove your claims):

1. $\mathcal{H}_{=k}^{\mathcal{X}} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| = k\}$. That is, the set of all functions that assign the value 1 to exactly k elements of $\mathcal{X}$.
2. $\mathcal{H}_{\text{at-most-}k}^{\mathcal{X}} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| \leq k \text{ or } |\{x : h(x) = 0\}| \leq k\}$.